E-12-CX-helicity-kill-test v1: does helicity give necessary CX and W_- link? Single artifact kill audit

ZFL Lab

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MISSION: Single kill test: does $T^* < \infty \Rightarrow C_H$ with C_H necessarily divergent and non-trivially linking to $W_- = \int (\omega^T S\omega)_-$ or to incompatibility $C_E^{weak} \wedge C_{L^3} \Rightarrow \perp$? No beautiful identity hunting.

INPUT: $\mathcal{C}_{PROVEN} = \{C_E^{weak} : \int W \to \infty,\ C_{L^3} : \|u\|_3 \to \infty\}$, G locked 0a37811, E-05 to E-09 BLOCKED, E-10 KILLED fast, E-11 CANDIDATE/OPEN without bridge.

FORBIDDEN: Beautiful helicity identity without W_- link. H bound $\Rightarrow W_-$ controlled. C_H not necessarily divergent under $T^* < \infty$. Indirect qualitative link to stretching. e_i, λ_i, η_- reuse. $C_E^{weak} \Rightarrow C_E^{strong}$.

AUDIT:

1. Helicity definitions:

$H = \int u \cdot \omega \, dx$, $\omega = \nabla \times u$, $\nabla \cdot u = 0$, periodic or $\mathbb{R}^3$ decaying.

Euler: $D_t H = 0$ for smooth (conservation). Navier-Stokes:

$\frac{d}{dt} H = -2\nu \int \nabla u : \nabla \omega \, dx = -2\nu \int \omega \cdot (\nabla \times \omega) \, dx$ (up to boundary). Exact form depends on BC but dissipative.

Estimate: $|dH/dt| \leq 2\nu \|\nabla u\|_2 \|\nabla \omega\|_2 \leq C\nu \|\omega\|_2 \|\nabla \omega\|_2$.

Thus $|H(t)| \leq \|u\|_2 \|\omega\|_2$.

2. Does $T^* < \infty \Rightarrow C_H$ with divergence?

If blow-up, enstrophy $\|\omega\|_2^2 \to \infty$ may occur (since $\int W \to \infty$ implies enstrophy divergence). Then $\|u\|_2$ bounded by initial energy (Leray). So $|H| \leq \|u\|_2 \|\omega\|_2$ may diverge as $\|\omega\|_2 \to \infty$, but not necessarily: u and ω could become orthogonal in L^2? H could stay bounded even if $\|\omega\|_2 \to \infty$ because alignment may cancel. No lower bound.

Thus no necessary divergence: $\boxed{T^* < \infty \not\Rightarrow \limsup |H| = \infty \text{ necessarily}}$.

At best $\int_0^{T_*} |dH/dt| = \infty$ may follow if $\int \|\omega\|_2 \|\nabla \omega\|_2 = \infty$, but $\int \|\nabla \omega\|_2^2$ may diverge under C_E^{weak} (since $\int W = \frac{1}{2}\|\omega\|_2^2 + \nu \int \|\nabla \omega\|_2^2$). So helicity variation may diverge, but this is consequence of C_E^{weak}, not new.

3. Link to W_-?

$W = \int \omega^T S \omega$, $H = \int u \cdot \omega$.

No pointwise relation. Betchov-type identity for H? None.

Write $W_- = \int |\omega|^2 (\xi^T S \xi)_-$, H contains u, not S. Using $u = (-\Delta)^{-1} \nabla \times \omega$ (Biot-Savart), $H = \int \omega \cdot (-\Delta)^{-1} \nabla \times \omega = \int |\omega|^2$ with nonlocal kernel. No sign control.

Try inequality: $W_- \leq |W| \leq C_S \|u\|_3 \|\nabla \omega\|_2^2$ from E-09, H does not improve constant nor sign.

Potential C_H: $\int_0^{T_*} |H| = \infty$? Not incompatible with $C_E^{weak} \wedge C_{L^3}$, since C_E^{weak} already says enstrophy-related quantities diverge. No contradiction:

$\boxed{C_E^{weak} \wedge C_{L^3} \wedge (\text{helicity bounds/divergence}) \not\Rightarrow \perp}$.

Thus fails kill criteria 2,3,4:

1. Only recovers helicity dissipation estimate $dH/dt = -2\nu \int \omega \cdot \operatorname{curl} \omega$, conservation in Euler. 2. Only produces bound $|H| \leq \|u\|_2 \|\omega\|_2$ or $|dH/dt| \leq C\|\omega\|_2 \|\nabla\omega\|_2$, no W_- control. 3. Helicity can remain bounded/compatible with blow-up (no necessary divergence). 4. No C_X necessarily divergent beyond PROVEN. 5. Connection to stretching only indirect via Biot-Savart, not W_-.

Therefore E-12 KILLED.

OUTPUT:

$$\boxed{H = \int u \cdot \omega,\ dH/dt = -2\nu \int \omega \cdot \operatorname{curl} \omega \text{ EXACT}}$$

$$\boxed{|H| \leq \|u\|_2 \|\omega\|_2,\ |dH/dt| \leq 2\nu \|\nabla u\|_2 \|\nabla\omega\|_2}$$

$$\boxed{T^* < \infty \not\Rightarrow |H| \to \infty \text{ necessarily; variation divergence already implied by } C_E^{weak}}$$

$$\boxed{\text{No } W_- \leq F(H), \text{ no } C_H \text{ giving } C_E^{weak} \wedge C_{L^3} \wedge C_H \Rightarrow \perp}$$

Classification: KILLED fast per rules 1-5.

$\mathcal{C}_{PROVEN} = \{C_E^{weak}, C_{L^3}\}$ intact.

E-10=KILLED, E-11=CANDIDATE/OPEN without bridge, E-12=KILLED.

Thus E-CX SEARCH = CLOSED/BLOCKED as package pressure, nabla xi, helicity produced no bridge.

Return to main diagram: $T^* < \infty \Rightarrow C_E^{weak}, C_{L^3}$, central question $C_E^{weak} \wedge C_{L^3} \Rightarrow \perp$ still OPEN.